# Hasan Kamran

*DevOps Engineer*

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## Profile

Cloud and Platform Engineer building Kubernetes platforms on AWS EKS and on-premises RKE2/Rancher, plus containerised workloads on ECS Fargate. Owns GitOps-driven, security-first infrastructure end to end with Terraform and CI/CD. AWS Solutions Architect (Associate) and Azure Administrator (AZ-104) certified.

## Skills

**Containers:** Kubernetes, Helm, Docker | **IaC:** Terraform | **CI/CD:** GitHub Actions, ArgoCD, OIDC

**AWS:** EKS, ECS, VPC, IAM, ALB, EC2, Route 53, Lambda, S3 | **Scripting:** Bash, PowerShell

**Security:** External Secrets Operator, AWS Secrets Manager, Trivy | **Azure:** Entra ID, Azure DevOps, RBAC

**Observability:** Prometheus, Grafana | **Linux Administration**

## Experience

**DevOps Engineer** 04/2026 – Present | UK

*BiP Solutions Ltd*

- Designed and built an on-prem **Kubernetes platform PoC** (RKE2, Rancher, vSphere) to production-grade patterns, solo, with no external consultancy, **taking per-app provisioning from 5 manual tasks to 0**.
- Configured MetalLB and nginx ingress for on-prem load balancing, giving services **stable external IPs and hostname-based routing with no cloud provider**.
- Automated DNS and TLS with ExternalDNS (Route 53) and cert-manager (Let's Encrypt), so **new apps get records and valid certificates with zero manual steps**.
- Deployed Longhorn storage with **3x replication across nodes**, so application data survives a node failure.
- Implemented GitOps CI/CD with Azure DevOps, Trivy and ArgoCD, taking changes from **commit to deployed in ~5 minutes** with **100% of images vulnerability scanned** before reaching the registry.
- **Eliminated static AWS credentials** using IAM Roles Anywhere and OIDC federation, cutting credential lifetime from **indefinite to 1 hour** and scoping **100% of IAM roles to a single identity**.
- Centralised application secrets in AWS Secrets Manager via External Secrets Operator, keeping **all sensitive values out of git**.
- Built monitoring for a critical, unchecked company DNS record (Prometheus, Grafana, systemd), **cutting detection from customer complaint to under 1 hour**.
- Replaced a planned hardware purchase with a serverless video rendering pipeline (Remotion, AWS Lambda, S3), **rendering in ~30 seconds at ~$0.03 per render** with no idle infrastructure.
- Led **Cyber Essentials Plus** certification end to end, driving remediation of **46 critical and high findings** and **closing 100%** across the in-scope estate.

**DevOps Engineer/Mentor (Open Source Community)** 09/2025 – Present | UK

*CoderCo*

- Designed and delivered containerised platforms on **AWS EKS** and **ECS** using **Terraform**, owning infrastructure from VPC design through to workload deployment.
- Implemented **GitOps-driven delivery** via **ArgoCD**, with automated TLS and DNS management using cert-manager and ExternalDNS.
- Built **GitHub Actions** CI/CD pipelines with integrated security scanning, reducing deployment steps from **5 to 1.**
- Secured pipeline authentication using **OIDC** with least-privilege IAM roles, eliminating long-lived credentials.
- Configured **IRSA** to scope AWS permissions to individual service accounts, **enforcing least privilege** at the pod level rather than the node.
- Mentor community members via live 1:1 calls, reviewing their infrastructure projects and helping troubleshoot errors and design decisions.

**IT Engineer (Automation & Security)** 08/2025 – 03/2026 | UK
*NHS Lanarkshire*

- **Automated vulnerability scans** across **~400 servers**, reducing manual overhead and improving consistency across the estate.
- Developed a **PowerShell script** to **automate server software updates** and strengthen security posture.
- Investigated **Microsoft Defender alerts** within a **SOC-style environment**, triaging and escalating security incidents across the organisation.
- Identified and resolved recurring false positive alerts triggered by **MDI sensors** using **KQL queries** to correlate flagged IPs against known internal devices.
- Analysed **suspicious URLs and attachments** reported by staff using a **sandbox environment**, assessing threat severity and determining appropriate response.
- Enforced **endpoint security policies** by identifying unauthorised software and driving remediation.
- Contributed to a **Windows 11 rollout** across multiple hospital sites, liaising with clinical staff to minimise disruption to critical services.

## Projects

**Self-Hosted Git Platform on AWS EKS**
*EKS · Terraform · ArgoCD · Helm · Traefik · cert-manager · ExternalDNS · Cloudflare · OIDC · IRSA · KMS · Prometheus · Grafana · Trivy · Checkov · GitHub Actions*

- Deployed a self-hosted **Gitea** platform on **AWS EKS**, provisioned end to end with modular **Terraform**
- Implemented **GitOps** via **ArgoCD**, ensuring every cluster change is version controlled and auditable
- Secured CI/CD with **GitHub Actions OIDC** and **IRSA**, eliminating stored credentials entirely
- Automated TLS and DNS with **cert-manager** and **ExternalDNS**, removing all manual certificate management
- Shift-left security scanning with **Trivy** and **Checkov** catching vulnerabilities before reaching the cluster

**Monitoring Platform on AWS ECS (Gatus)**
*ECS Fargate · Terraform · GitHub Actions · IAM · ALB · CloudWatch · SSM*

- Deployed **Gatus** monitoring platform on **ECS Fargate** using **Terraform** with immutable container deployments
- Implemented **OIDC**-enabled CI/CD with least-privilege IAM, eliminating reliance on long-lived credentials
- Configured private networking behind an **ALB**, keeping workloads off the public internet
- Centralised container logs via **CloudWatch**, enabling faster troubleshooting and operational visibility

**AWS IAM Auditing Tool**
*Bash · AWS CLI · IAM · S3 · SNS*

- Built a **Bash** tool to detect over-privileged IAM assignments across users, roles, and groups
- Triggers **SNS alerts** on violations and uploads reports to **S3** for audit trail and governance

## Certifications & Recognition

AWS Certified Solutions Architect – Associate | AWS Certified Cloud Practitioner

Microsoft Azure Administrator (AZ-104) | Microsoft Security Fundamentals (SC-900)

AWS All Builders Welcome Grant — re:Invent 2026

## Education

**CertHE Civil Engineering with Architecture**
*University of Glasgow*